Market Sentiment & Trader Behavior Analysis

Executive Summary

This project investigates the relationship between market sentiment regimes (Fear vs. Greed) and trader behavior using two complementary datasets: detailed trader-level transaction records and a daily market sentiment index. The primary objective was to evaluate how sentiment conditions influence trading performance, risk-taking behavior, and consistency across different trader segments.

Methodology

Trades were aggregated at a daily, per-account level and aligned with corresponding daily sentiment classifications. Key performance and behavioral metrics were engineered, including daily Profit & Loss (PnL), trade frequency, average trade size, and long/short exposure ratios. Traders were segmented based on activity level, position size, and PnL consistency to assess whether sentiment effects vary across behavioral profiles. This segmentation enabled deeper insights into risk tolerance, adaptability, and performance stability under different market regimes. An additional exploratory predictive modeling component was conducted to evaluate whether sentiment and behavioral indicators could forecast next-day trader profitability.

Key Insights

- Performance varies across sentiment regimes: Greed periods show higher average profitability but also elevated volatility, while Fear periods demonstrate more conservative yet stable outcomes.
- Behavioral shifts are sentiment-driven: Traders increase trade frequency and position sizes during Greed regimes, indicating heightened risk appetite.
- Segment-level differences are significant: Consistent traders maintain relatively stable performance across regimes, whereas high-frequency and inconsistent traders experience amplified swings in profitability.

Strategy Recommendations

- Risk Mitigation During Fear Regimes: High-frequency and inconsistent traders should reduce trade frequency and scale down position sizes to manage downside risk exposure.
- Selective Expansion During Greed Regimes: Consistent traders may cautiously increase engagement to capture momentum opportunities while maintaining disciplined risk controls.

- Behavior-Based Risk Controls: Adaptive position sizing and dynamic exposure management should be aligned with both sentiment regime and trader profile.

Predictive Modeling (Bonus Analysis)

A logistic regression model was developed to predict next-day trader profitability using sentiment indicators and engineered behavioral features. Although predictive performance was constrained by class imbalance and market noise, the exercise reinforced key principles in financial modeling: careful feature selection, avoidance of overfitting, and realistic evaluation using appropriate performance metrics. The modeling exercise demonstrates that while sentiment provides signal value, robust predictive systems require richer feature engineering and regime-aware validation.